MILITARY SPECIFICATION

RESIN, CASTING, EPOXY, ALUMINUM FILLED

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope.- This specification describes the requirements of the procurement of one type of aluminum-filled, epoxy, casting resin. The resin is a 100 percent solid, two-component, aluminum-filled, low-viscosity compound suitable for forming plastic dowels for alignment purposes. The resin cures at room temperature.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, or such portion thereof as may be designated herein, of the issue in effect on the date of invitation of bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

L-P-406	Plastic, Organic: General Specifications, Test Methods
QQ-M-44	Magnesium Alloy Plate and Sheet (AZ31)
QQ-S-763	Steel Bars, Shapes, and Forgings, Corrosion Resisting
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood Nailed and Lockcorner
PPP-B-636	Boxes, Fiber

FSC 9330

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MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-W-21927	Weapons, Handling and Preparation for Delivery of; General Requirements for
MIL-M-45202	Magnesium Alloys, Anodic Treatment of

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-171	Systems for Preparation, Painting, and Finishing Metal and Wood Surfaces

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

NAVWEPS OD 12328	Instructions for Preparing and Coating Plastic Bushings

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS PUBLICATION

ASTM D1002	Method of Test for Strength Properties of Adhesives in Shear-by-Tension Loading (Metal-to-Metal)

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(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

UNIFORM CLASSIFICATION COMMITTEE PUBLICATION

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Ill.)

AMERICAN TRUCKING ASSOCIATION, INC. PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1242 16th Street, N.W., Washington 6, D. C.)

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Pre-production samples. - Unless otherwise specified in the contract or order, pre-production samples of the kit or unit (see 5.1) shall be manufactured by the methods and procedures proposed for the production lot. These samples shall be tested as specified in section 4 herein and are for the purpose of determining that the production items meet the requirements of this specification and the design. See 4.1.1.

3.1.2 Materials. - The resin shall consist of an epoxy containing aluminum powder of not less than 99 percent purity. The size of the aluminum powder particles shall be such that 100 percent of the powder will pass through a 100-mesh sieve, and at least 75 percent will pass through a 325-mesh sieve. The materials composing the casting resin system shall be such that the requirements of this specification are met.

3.1.3 Requests for deviation. - Each request that a deviation be authorized shall have been submitted to the Procuring Activity, and the deviation shall have been authorized by the Bureau of Naval Weapons or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the Procuring Activity to evaluate its effect on performance and reliability and the reasons for the suggested

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deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder) and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.

3.1.4 Conflicting requirements. - Conflicting requirements arising between this specification and any specification, publication, or drawing listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.2 Definitions. - Not applicable.

3.3 Performance requirements and product characteristics. - The casting resin shall meet the following performance requirements and product characteristics:

3.3.1 Physical and chemical properties.

3.3.1.1 Appearance.

3.3.1.1.1 Uncatalyzed resin. - The uncatalyzed resin shall be free from agglomerated filler, and in a smooth homogeneous condition. Slight settling of the filler is permitted, provided it can be mixed back into a homogeneous mixture, free of agglomerates. See 4.4.4.1.1.

3.3.1.1.2 Catalyst. - The catalyst shall be a clear, transparent liquid of water-like viscosity. See 4.4.4.1.2.

3.3.1.1.3 Casting. - Bubbles and pinholes, if present in a casting of the catalyzed, cured resin, shall be small, and shall not be numerous. See 4.5.1.

3.3.1.2 Viscosity. - The viscosity of a 35-gram sample of a 50-gram batch of the catalyzed resin shall be 1200 plus or minus 150 centipoises after 30 minutes. See 4.5.2.

3.3.1.3 Gel time. - The gel time of a 50-gram batch of the catalyzed resin shall be 67 plus or minus 12 minutes. See 4.5.3.

3.3.1.4 Corrosion to metals. - The resin shall not cause any visible corrosion to the metals listed below when specimens of these metals, finished as specified below, and coated with the catalyzed resin, are subjected to 15 cycles of high humidity in accordance with procedure I of Specification MIL-E-5272. See 4.5.4.

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(a) Magnesium, alloy AZ31B, Specification QQ-M-44; anodized in accordance with the type II, Class D process of Specification MIL-M-45202.

(b) Steel, Class 303, condition A, cold finished, Specification QQ-S-763; passivated in accordance with finish 5.4 of Standard MIL-STD-171.

3.3.1.5 Hardness. - The hardness of the catalyzed resin after a 22 to 26 hour cure at room temperature shall be a minimum of 101 when measured with a Rockwell tester (R scale). See 4.5.5.

3.3.1.6 Shear strength. - The catalyzed resin shall have a shear strength of not less than 3000 pounds per square inch. See 4.5.6.

3.3.1.7 Adhesion strength. - The catalyzed resin shall have an adhesion strength of not less than 2750 pounds per square inch. See 4.5.7.

3.4 Environmental requirements.

3.4.1 Storage stability. - All the requirements of this specification shall be met after the catalyst and uncatalyzed resin, packaged in accordance with the requirements of 5.1, are stored for 6 months minimum under the environmental conditions specified in 4.4.2. See 4.5.8.

3.4.2 Shock-transportation, storage and handling. - The complete pack, after being subjected to the tests specified in 4.5.9 shall not rupture so as to spill contents or break seals on unit packages.

3.5 Workmanship. - The workmanship shall be in accordance with high grade commercial practice covering this type of material.

3.5.1 The workmanship exhibited in pre-production samples will be evaluated to determine acceptability. The approved standards of workmanship will thereby become a minimum production requirement for all units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Pre-production sample. - Unless otherwise specified for each contract or order, a representative sample of three kits or units shall be manufactured by the methods and procedures proposed for the production lot. These shall be submitted for pre-production tests at an activity designated by the Bureau of Naval Weapons, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the pre-production samples shall be at the contractor's risk. Pre-production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

4.1.2 Periodic production samples. - Unless otherwise specified by the procuring agency the tests of 4.4 and 4.5 shall be evaluated as follows:

Step 1. - Upon initiation of production, select one (1) unit from each group of ten (10) units for submission to an activity designated by the procuring agency.

Step 2. - Upon satisfactory compliance of each unit from three (3) consecutive groups, select one (1) unit from each group of twenty-five (25).

Step 3. - Upon satisfactory compliance of each unit from three (3) consecutive groups, select one (1) unit from each group of fifty (50) units. Continue this procedure until a defective unit is found.

Step 4. - When a defective unit is found in any group, the group represented by the sample shall be rejected. Shift to step 1 and proceed as indicated.

4.2 Classification of test. - The inspection and testing of the kits or unit shall be classified as follows:

4.2.1 Acceptance tests. - These tests shall be accomplished on each kit or unit submitted for acceptance under contract. These tests are detailed in 4.4.

4.2.2 Pre-production and periodic production tests. - Pre-production and periodic production tests shall be accomplished on samples selected as specified in 4.1.1 and 4.1.2 which are representative of the production of the kit or unit after the award of the contract, to determine that the

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production meets the requirements of this specification. These tests are detailed in 4.5. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with these requirements will result in the rejection of the lot or the cessation of production, as determined by the procuring activity.

4.2.3 Tests after delivery. - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all the tests specified herein, as deemed necessary, after installation of the kit or unit. Any defects discovered during or after these tests due to the contractor's failure to comply with drawings or specifications for this equipment shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.3 Test equipment. - No test equipment is required to perform the acceptance tests set forth in this specification.

4.4 Acceptance tests.

4.4.1 The contractor shall conduct acceptance tests as specified in 4.2.1 to assure that the kit or unit is in compliance with the requirements of 3.1, 3.3.1.1.1, 3.3.1.1.2, and 3.5.

4.4.2 Test conditions. - Unless otherwise specified, the kit or unit shall be subjected to acceptance tests under the following conditions:

4.4.2.1 Temperature: - Room ambient 65 to 95 degrees Fahrenheit (18 to 35 degrees centigrade).

4.4.2.2 Altitude: - Normal ground atmospheric pressure.

4.4.2.3 Vibration: - None.

4.4.2.4 Humidity: - Room ambient to 95 percent relative maximum.

4.4.3 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary facilities and personnel for performing all acceptance tests.

4.4.4 Test procedure. - The detailed test procedure listed in the following paragraphs shall be followed.

4.4.4.1 Appearance.

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4.4.4.1.1 Uncatalyzed resin. - The appearance of the uncatalyzed resin shall be checked visually to determine compliance with the requirements of 3.3.1.1.1.

4.4.4.1.2 Catalyst. - The appearance of the catalyst shall be checked visually to determine compliance with the requirements of 3.3.1.1.2.

4.4.5 Preservation, packing and packaging. - Preservation, packing and packaging of the aluminum-filled, epoxy casting resin shall conform to section 5 of this specification.

4.5 Pre-production and periodic production sample tests.

4.5.1 Appearance. - A specimen casting shall be prepared and shall be inspected visually to determine compliance with the requirements of 3.3.1.1.3. Preparation of the casting resin, and the curing time and temperature shall be in accordance with the application information specified in OD 12328. The casting shall be formed by allowing about 25 grams of catalyzed resin to cure in an aluminum weighing cup. In the case of kits, the contents of the containers of resin shall be combined before adding the catalyst. Care shall be taken not to trap air in the catalyzed resin if it is poured into the aluminum weighing cup from another container.

4.5.2 Viscosity. - The viscosity of a 35 plus or minus 1 gram sample of a freshly made 50-gram batch of the catalyzed resin shall be measured in a glass test tube having an outside diameter of 1 inch and a length of 3 inches to determine compliance with the requirements of 3.3.1.2. A viscometer, Brookfield Engineering Laboratories Model RVF with a number 2 LVT spindle operated at 20 revolutions per minute, or equivalent, shall be used to measure the viscosity of the resin, and the temperature of the resin shall be maintained at 25 plus or minus 2 degrees centigrade.

4.5.3 Gel time. - The gel time of a freshly made 50-gram batch of the catalyzed resin shall be measured in a glass test tube having an outside diameter of 2.5 centimeters and a length of 15 centimeters to determine compliance with the requirements of 3.3.1.3. A gel-time meter, Sunshine Scientific Company - catalog number 22, General Electric Company - catalog number 9125531G1, or equivalent, shall be used to measure the gel time of the resin, and the temperature of the resin shall be maintained at 25 plus or minus 2 degrees centigrade.

4.5.4 Corrosion to metals. - Panels of each of the metals listed in 3.3.1.4, approximately 4 inches square and at least 0.050 inch thick, and finished as specified in 3.3.1.4, shall be coated with the catalyzed resin for a coating thickness of 0.01 to 0.02 inch. The resin may be applied by mounting the panel horizontally with a barrier around its edges, and pouring

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in the appropriate amount of freshly catalyzed resin, or by dipping the panel into or brushing on the catalyzed resin as it approaches gelation. The coating thickness shall be determined by measuring the thickness of the panels before and after application and curing of the coating.

The coated panels shall be cured under the conditions specified in 4.4.2 for 22 to 26 hours, and then subjected to humidity cycling as specified in 3.3.1.4. At the end of the humidity cycling, the coated area of each panel shall be inspected visually to determine compliance with the requirements of 3.3.1.4.

4.5.5 Hardness. - A resin casting, not smaller than 1 inch by 1 inch by 1/2 inch shall be prepared, and cured 22 to 26 hours under the conditions of 4.4.2. After curing, the hardness of the speciemen shall be measured to determine compliance with the requirements of 3.3.1.5. A hardness tester, Wilson Mechanical Instrument Company - Rockwell Hardness Tester, or equivalent, shall be used to measure the hardness of the specimen casting in accordance with method 1081.1 of Specification L-P-406. The casting can be prepared in an aluminum weighing cup.

4.5.6 Shear strength.

4.5.6.1 Preparation of specimen casting. - A specimen of cast resin shall be prepared as follows:

(a) Three cold rolled steel panels, each 5 inches long by 1/2 inch wide by 1/4 inch thick, shall be assembled as shown in figure 1,

(b) A 0.250 + .005 - .002 inch diameter hole shall be drilled through the overlapping areas of the panels with the center of the hole passing through the center line of each panel,

(c) Additional jig aligning holes shall be drilled at opposite ends of the panels,

(d) The catalyzed resin shall be poured into the hole through the overlapping areas of the panels to fill it completely, maintaining a constant rate of flow and a small jet diameter to avoid trapping air,

(e) The resin casting so formed shall be cured for 22 to 26 hours under the conditions specified in 4.4.2.

4.5.6.2 Conduct of test. - The specimen of cast resin shall be tested in accordance with ASTM test method D 1002 to determine compliance with the requirements of 3.3.1.6, using a Widemann Machine Company or Baldwin-Lima-Hamilton Corporation, Baldwin-Emery-Tate 1200/12,000/60,000-pound universal testing machine, or equivalent.

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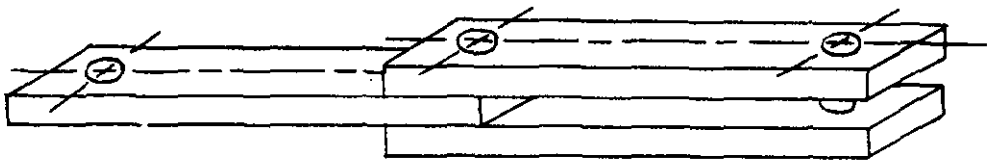


Figure 1. Test Fixture for Shear Strength Test of Resin Casting

4.5.7 Adhesion strength.

4.5.7.1 Preparation of specimen casting.- A specimen of cast resin shall be prepared as follows:

(a) A 0.250 +.005 -.002 inch diameter hole shall be drilled through the center of a 2 by 2 inch face of a 2 by 2 by 1/2 inch block of magnesium,

(b) The inside diameter of the hole, and other surfaces of the block if, desired, shall be anodized using the anodizing process specified in 3.3.1.4(a),

(c) A straight 0.630 +0.001 -0.001 inch diameter by 9/16 inch minimum length dowel pin of magnesium or corrosion-resistant steel, coated with a suitable mold release compound such as Chemical Development Company's Mold Release B, shall be inserted into the hole and located concentrically so as to provide a clearance of about 0.093 inch all around the pin,

(d) The catalyzed resin shall be injected into the cylindrical space around the dowel pin by means of a hypodermic needle having a 0.020 to 0.030 inch diameter orifice, completely filling the space, without air bubbles,

(e) The resin bushing so formed shall be cured for 112 to 128 hours under the conditions specified in 4.4.2, and dowel pin shall be removed at the end of the curing period.

4.5.7.2 Conduct of test.- Using the testing machine specified in 4.5.6, a metal ram 0.2490 to 0.2495 inch in diameter, and other test

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fixtures as required, the force required to push the resin bushing out of the magnesium block shall be measured, applying force at a rate of 600 to 800 pounds per minute to determine compliance with the requirements of 3.3.1.7 (for compliance with the requirements of 3.3.1.7, the force required shall be 1080 pounds minimum).

4.5.8 Storage stability. - The kit or unit, packaged as specified in 5.1, shall be stored as specified in 3.4.1, and at the end of the storage period shall be tested to determine compliance with the requirements of 3.1, 3.3, and 3.5.

4.5.9 Shock-transportation, storage and handling. - The complete pack, prepared as specified in section 5 shall be subjected to the appropriate rough handling tests specified in MIL-W-21927.

4.6 Functional tests. - At the conclusion of the tests specified in 4.5.9 and other tests, as applicable, shall be conducted on the unit packages to determine freedom from rupture of seal and compliance with all the requirements of section 3.

4.7 Rejection. - Failure of the kit or unit to meet any of the applicable requirements of this specification shall be considered cause for rejection.

4.8 Waiver of tests. - Tests specified in 4.5.9 may be waived by the procuring agency for level C packs only, provided the contractor guarantees safe delivery to the using activity.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning, drying and preservative application. - Not applicable.

5.1.1.2 Unit packaging.

5.1.1.2.1 Unit packaging kits. - Unit packaging of the resin for kits shall be in accordance with Method IA-13 of Specification MIL-P-116. The unit package shall contain 10 plus or minus 0.5 grams of resin packaged in a clean, wide-mouthed glass jar. Unit packaging of the catalyst for kits shall be in accordance with Method IA-13 of Specification MIL-P-116. The unit package shall contain sufficient catalyst to catalyze six 10-gram jars of resin packaged in a single glass bottle.

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5.1.1.2.2 Unit packaging, units. - Unit packaging of the resin for units shall be in accordance with Method IA-5 of Specification MIL-P-116. The unit package shall contain either 25 grams, plus or minus 1 gram, 50 grams plus or minus 1 gram or 200 grams, plus or minus 2 grams of resin packaged in a wide-mouthed container. Unit packaging of the catalyst for units shall be in accordance with Method IA-13 of Specification MIL-P-116. The unit package shall contain the proper amount of catalyst to catalyze 25 grams, 50 grams or 200 grams of resin respectively packaged in clean glass jars.

5.1.1.3 Intermediate package.

5.1.1.3.1 Intermediate package, kits. - Six jars of resin plus one bottle of catalyst packaged as described in 5.1.1.2.1 and an eye dropper constitute a kit. The kits shall be intermediately packaged in fiber boxes in accordance with Specification PPP-B-636, Class 2. The intermediate package shall contain one (1) kit.

5.1.1.3.2 Intermediate package, units. - Not applicable.

5.1.1.4 Cushioning. - The unit package, intermediate package and pack, shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices, to ensure that the kits or units will meet all the requirements of section 3 after being tested in accordance with 4.

At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent. The pH range of the material shall be between 6.0 and 8.0.

5.1.2 Level B. - Not applicable.

5.1.3 Level C.

5.1.3.1 Cleaning, drying and preservative application. - Not applicable.

5.1.3.2 Unit packaging.

5.1.3.2.1 Unit packaging, kits. - The unit package of the resin for kits shall contain 10 plus or minus 0.5 grams of resin packaged in a clean wide-mouthed glass jar. The unit package of the catalyst for kits shall contain sufficient catalyst to catalyze six 10-gram jars of resin packaged in a single glass bottle. The unit packaging of the resin and the catalyst shall be in accordance with the manufacturer's commercial practice, when adequate.

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5.1.3.2.2 Unit packaging, units. - The unit package of the resin for units shall contain either 25 grams, plus or minus 1 gram, 50 grams plus or minus 1 gram or 200 grams plus or minus 2 grams of resin packaged in a wide-mouthed container. The unit package of the catalyst for units shall contain the proper amount of catalyst to catalyze 25 grams, 50 grams or 200 grams of resin respectively packaged in clean glass jars. The unit packaging of the resin and the catalyst shall be in accordance with the manufacturer's commercial practice, when adequate.

5.1.3.3 Intermediate package.

5.1.3.3.1 Intermediate package, kits. - Six jars of resin plus one (1) bottle of catalyst packaged as described in 5.1.3.2.1 and an eye dropper constitute a kit. The kit shall be intermediately packaged individually in accordance with the manufacturer's commercial practice, when adequate.

5.1.3.3.2 Intermediate package, units. - Not applicable.

5.2 Packing. - Unless otherwise specified in the contract or purchase order, the number of packages per pack shall be in accordance with table I.

5.2.1 Level A.

5.2.1.1 Exterior containers. - Kits or units packaged as described in 5.1.1.2.1 or 5.1.1.2.2 shall be packed in a wood box in accordance with Specification PPP-B-601 overseas type or PPP-B-621, class 2 (gross weight not to exceed 150 lbs) or in fiber boxes in accordance with Specification PPP-B-636, class 2 (gross weight not to exceed 70 lbs).

5.2.2 Level B.

5.2.2.1 Exterior containers. - Kits or units packaged as described in 5.1.1.2 or 5.1.1.2.2 shall be packed in a wood box in accordance with Specification PPP-B-601, domestic type or PPP-B-621, class 1 (gross weight not to exceed 150 lbs) or in fiber boxes in accordance with Specification PPP-B-636, class 1 (gross weight not to exceed 70 lbs).

5.2.3 Level C. - Kits or units packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

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Table I. Number of Packages Per Box

Package	Quantity Per Box
Kit	22
Resin, 25-grams	50
Catalyst for 25 grams of resin	100
Resin, 50-grams	24
Catalyst for 50 grams of resin	100
Resin, 200-grams	24
Catalyst for 200 grams of resin	50

5.3 Marking.

5.3.1 Special markings.

5.3.1.1 Special markings, kits. - The jars of resin shall be marked to identify and give the nominal weight of the contents. The bottle of catalyst shall be marked to indicate the number of drops of catalyst to be added to each 10 grams jar of resin. Each kit shall contain mixing and curing instructions, and the shelf life and date of manufacture.

5.3.1.2 Special markings, units. - The containers shall be marked to identify and give the nominal weight of the contents. Each unit shall contain mixing and curing instructions, and the shelf life and date of manufacture.

5.3.1.3 Manufacturer and batch markings.

5.3.1.3.1 Packages. - Each package shall be marked with the name of the manufacturer, the batch or lot number, and the date of manufacture.

5.3.1.3.2 Boxes. - Each box shall be marked with the name of the manufacturer, and the batch or lot numbers of the packages contained therein.

5.3.2 Normal markings. - In addition to the markings required by contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. - The casting resin is intended for use in the Angle Solvers Mark 18 Mod 2 and Mod 3, the Fire Control System Mark 112 Mod 1, the Sonar Receiver Sets AN/BQG-1, -2, -3, -4, or other suitable application.

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6.2 Ordering data. - Procurement documents should specify the specific title, number, and date of this specification and exceptions to this specification; applicable drawings and other documents; selections of applicable levels of packing; and guarantee clause. (See 6.5.)

6.2.1 The attention of the contracting officer is called to 5.2.

6.3 Criteria for use of the proper level of preservation, packaging and packing shall be as follows:

For Level A. - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. - This level shall be used only when it is definitely known that the item will be held in covered storage, overseas locations, for 6 months or less, or in domestic locations for an indefinite period.

For Level C. - This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116 and to paragraph 4.8.

6.5 Guarantee. - The contract or order shall include a requirement for a guarantee stating that any kit or unit which fails to meet the requirements of this specification within six months after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

6.6 The contract or order shall include a sufficient number of extra items to cover production and pre-production testing samples.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented inventions that may in any way be related thereto.

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DEPARTMENT OF THE NAVY
BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
NAVY DEPARTMENT

Chief, Bureau of Naval Weapons (RREN-5)

Navy Department

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 48-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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